# A Mutually Enhancement Network for Superpixel Segmentation and Classification of Hyperspectral Image

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***Abstract*—Most existing hyperspectral image (HSI) classification methods primarily focus on capturing subtle spectral variations by leveraging local spectral–spatial cues derived from patch-level representations. However, limited attention has been given to exploring the global spatial contextual correlations among pixels of HSI. In this study, we propose the superpixel segmentation and classification mutual enhancement network ($S^2$CMEN), a novel framework that integrates global spatial correlations with spectral information through the mutual enhancement of superpixel segmentation and classification. Specifically, a global spatial adaptive module (GSAM) is designed to obtain the direct correlation of the global classes in HSI. It consists of an adaptive spectral-superpixel network (ASSN) and a graph convolutional network (GCN), forming a synergistic architecture that effectively captures global spatial relationships by adaptively deriving superpixel results from HSIs. Notably, GSAM offers a transferable global spatial representation for HSI tasks, enabling integration with other spectral feature extraction models. Furthermore, we develop a spatial–spectral fusion module (SSFM) to obtain comprehensive spectral features and fuse them with the extracted global spatial features. Finally, under the constraint of a unit loss, the mutual enhancement strategy (MES) can make the superpixel segmentation loss and the classification loss mutually enhance each other for better performance. We conducted extensive experiments on three public datasets. The proposed $S^2$CMEN achieves overall classification accuracies of 97.38%, 92.33%, and 91.38% on Indian Pines (IPs), Pavia University (PU), and Houston, respectively, consistently surpassing existing state-of-the-art methods.**

***Index Terms*—Feature fusion, hyperspectral image (HSI) classification, superpixel segmentation, Transformer.**

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## I. Introduction

HYPERSPECTRAL image (HSI) captures both spectral and spatial information of a scene by simultaneously imaging the target features across successive bands [1], [2]. One of the critical tasks in HSI is classification, which involves assigning a label to each pixel in the image. Consequently, it finds extensive applications in remote sensing image analysis, including precision agriculture [3], environmental analysis [4], military surveillance [5], mineral exploration [6], and urban survey [7].

Over the past decade, various pattern recognition methods have been developed for feature extraction in HSI classification tasks, such as neural networks [8], support vector machines (SVMs) [9], broad learning system [10], and dynamic or random subspaces [11]. However, these approaches predominantly focus on pixel-based feature extraction. To integrate spatial context into pixel classifiers, deep learning (DL) has made significant breakthroughs in HSI classification tasks. DL excels in capturing both local spatial and spectral information [12]. Recently, the Transformer's multihead self-attention (MSA) mechanism has demonstrated efficacy in capturing long-range spectral information in HSI by aligning position codes with data [13], [14]. These methods have achieved success across various fronts.

The common practice of existing HSI classification methods segments the image into uniformly sized patches based on a "label," ensuring adequate spectral information extraction while retaining spatial context within each label [15], [16]. However, they often overlook global spatial information acquisition. Given that HSI constitutes a 3-D data cube, where the first two dimensions represent spatial dimensions, fostering interconnections among spatial information is equally crucial for enhancing classification performance [17], [18], [19]. Graph convolutional network (GCN) architectures can facilitate conveying global spatial information by modeling the similarity of entire image vectors [20], [21]. Nevertheless, they exhibit two limitations.

1) These GCN methods often incur high computational cost due to the requirement of constructing a large adjacency matrix [22], [23]. This problem is more serious in large-size HSI input.
2) As the depth of the network increases, the expressive power of GCN is diminished by transitional smoothing

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[24]. Together with the fact that GCN is overly concerned with spatial information, it also encounters bottlenecks when extracting spectral information.

For the first problem, the introduction of a superpixel segmentation process provides an effective solution. This technique aggregates pixels with similar textures, reducing their number to a smaller set of irregular superpixels that hold visual significance [25]. By expressing visual features through superpixels, the computational complexity associated with modeling spatial information is significantly reduced [26]. However, conventional superpixel segmentation methods rely on handcrafted features and lack adaptability to task-specific feedback. DL-based approaches for superpixel segmentation struggle to fully exploit the rich spectral information inherent in HSI as clustering criteria, as they typically utilize only basic image representations such as color and luminance. Regarding the second challenge, although acquiring spectral information through GCN is challenging, synergizing GCN with convolutional neural network (CNN)-based methods can effectively learn local spectral features by accommodating scale variations in large-size images [27]. Nonetheless, CNNs tend to overly emphasize local spatial information, overlooking the broader correlation between the entirety and its constituents, thereby potentially distorting global spectral information [28]. Additionally, the frequent size changes induced by multiple pooling layers pose challenges for the network in capturing detailed spectral information. Transformer-based approaches excel as global feature extractors for handling long-range spectral information [29]. However, when processing HSI as a whole image instead of segmented patches, Transformers incur high computational costs. Furthermore, when confronted with a plethora of classified classes exhibiting extremely similar spectral features, computing attention weights for the entire image may compromise the model's ability to capture local spectral features.

To address the aforementioned challenges, we propose the superpixel segmentation and classification mutual enhancement network ($\mathbf{S}^2$CMEN), which captures global spatial features with long-range spectral dependencies through the synergistic interaction between superpixel segmentation and classification. This mutual enhancement enables $\mathbf{S}^2$CMEN to effectively exploit both spatial and spectral information. The network comprises three key modules: the global spatial adaptive module (GSAM), the spatial–spectral fusion module (SSFM), and the mutual enhancement strategy (MES). Specifically, in the GSAM, it includes an adaptive spectral-superpixel network (ASSN) and a GCN. Unlike conventional superpixel segmentation methods, the designed ASSN is capable of leveraging spectral information to aggregate superpixels and adaptively generate high-quality superpixel results. Subsequently, global spatial features are captured by GCN. The GSAM can also complement other HSI feature extractors by providing additional global spatial information. For the SSFM, it has a Spectral-Swin Transformer and a fusion process for HSI classification. The Spectral-Swin Transformer employs a window-shifting attention mechanism, which partitions large-scale HSIs into multiple windows and computes attention separately within each. By shifting the windows, it further enables the effective extraction of long-range spectral features, particularly in scenarios with densely distributed classes. In SSFM, the derived global spatial features and long-range detail spectral features are combined for classification. Finally, in the MES, the superpixel segmentation loss and the classification loss mutually enhance each other under the constraint of a unit loss mechanism, yielding superior results for the entire $\mathbf{S}^2$CMEN. The main contributions of this article can be summarized as follows.

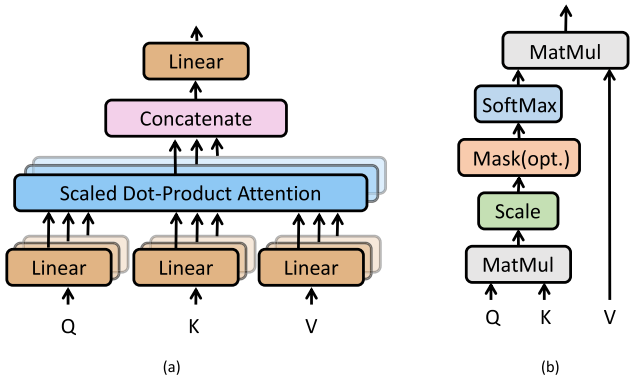



Fig. 1. Illustration of the attention mechanism in the Transformer. (a) MSA. (b) Self-attention.

1) We design the GSAM, which is a plug-in to capture global spatial information, addressing the deficiency in global spatial correlation in HSIs and thoroughly exploring the role of global spatial information in HSI classification tasks.
2) We design the Spectral-Swin Transformer within the SSFM to enable feature extraction from entire HSI inputs. By integrating convolutional layers with a window-shifting attention mechanism, the module captures fine-grained, long-range detail spectral features.
3) We propose a novel MES that establishes a mutual interaction between superpixel segmentation and classification. Feedback from the classification branch adaptively guides superpixel generation, while the refined segmentation in turn boosts classification accuracy, creating a synergistic improvement in both tasks.

The remainder of this article is organized as follows. Section II reviews the related work, while Section III provides a detailed description of our proposed $\mathbf{S}^2$CMEN. Section IV presents comprehensive experiments on $\mathbf{S}^2$CMEN to validate the effectiveness. Finally, Section V concludes this article.

## II. Related Work

### A. Transformer-Based Backbone in HSI Classification

The Transformer, as one of the most recent DL methods, was initially used in natural language processing before being adapted for computer vision tasks [30]. Its key advantage over CNNs is its ability to capture global contextual information through self-attention, allowing it to model long-range dependencies and extract more powerful features [31]. The structure of the Transformer is shown in Fig. 1. For each input element, it generates query (Q), key (K), and value (V) vectors through

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three learned matrices. The dot product between an element's Q vectors and the K vectors of other elements determines their correlation. These dot products are then scaled and passed through a softmax function. The resulting weights are applied to the value vectors, emphasizing elements with high attention scores. This self-alignment process enables tokens to gather information from each other effectively [32].

In order to apply Transformers to image processing, several approximation methods have been proposed in recent years. Liu et al. [13] designed a Swin Transformer architecture that can extract local features. It used nonoverlapping windows to split the mapped features and apply interwindow and cross-window interactions to construct hierarchical attention. This kind of local multipoint product self-attention block can completely replace convolution. Zhu et al. [33] proposed a deformable DETR model with a deformable CNN. Similarly, in [34], a sparse Transformer was introduced for text, graphics, and speech, capable of analyzing sequences up to 30 times longer than earlier attention-based algorithms. These special attention architectures have obtained good results in computer vision tasks [35], [36].

For HSI, the high dimensionality of the spectral information and the large interspectral differences require high detail processing of the model. Spectral information in HSI is complex, and there is high spectral variability among classes. The single-stage Transformer is difficult to achieve satisfactory training models. The emergence of Swin Transformer can solve the problems of high computational effort and poor dense target detection. However, they ignore the ability to extract global features, which prevents the Transformer from taking full advantage.

### B. Superpixel Segmentation

A superpixel comprises a group of adjacent pixels that share similar characteristics, such as color, brightness, texture, and so on, forming a small region of the image block [37]. Most of these small regions preserve valid information for further image segmentation and generally maintain the boundary integrity of objects in the image [38]. By "aggregating" pixels with similar characteristics, the image is compressed to form a more representative superpixel. One of the functions of superpixels is to act as a preprocessor for other algorithms in image processing, reducing complexity without sacrificing much accuracy.

Currently, only a tiny fraction of superpixel segmentation networks have combined them with deep neural networks. A major reason for this is that standard convolution operations are defined on regular grids and are inefficient when applied to superpixels. Existing superpixel algorithms are based on nearest neighbor operations, which are nondifferentiable [39]. However, in the process of neural network training, backpropagation algorithms are required to be used, and neural networks are difficult to train if a nondifferentiable module is introduced. Therefore, it is difficult to introduce superpixel segmentation in deep networks [40].

Some superpixel segmentation networks combined with DL make it difficult to integrate spectral information into the process of superpixel generation. Using only the base image representation, it is difficult to fully extract the global spatial information in HSI.

## III. Proposed Method

### A. Overall Architecture

The proposed $\mathbf{S}^2$CMEN processes HSIs by passing spatial and spectral information through three sequential modules: the GSAM, the SSFM, and the MES, as illustrated in Fig. 2.

The network begins with the GSAM, where the ASSN extracts spectral and spatial edge features, performs principal component analysis (PCA)-based dimensionality reduction, and generates superpixel centers and pixel–superpixel associations. These are used to construct graph nodes and the corresponding adjacency matrix, which the GCN processes to obtain global spatial representations.

The resulting spatial features are forwarded to the SSFM, which simultaneously receives spectral features extracted by the Spectral-Swin Transformer. The Spectral-Swin Transformer operates on the entire HSI using convolution layers and a window-shifting attention mechanism, producing long-range, fine-grained spectral representations. Spatial and spectral features are then fused to form a joint representation.

Finally, the MES stage jointly optimizes the classification and superpixel segmentation branches. Classification results provide feedback to refine superpixel generation, while improved segmentation benefits classification, forming a closed-loop enhancement process.

### B. Global Spatial Adaptive Module

In GSAM, the ASSN and GCN work in tandem; the ASSN reduces computational complexity and efficiently constructs nodes through superpixel segmentation, while the GCN leverages these results to model node correlations via an embedded graph representation, thereby producing global spatial features that enhance HSI classification performance. The detailed working principles of the ASSN and GCN are presented in the following.

*1) Adaptive Spectral-Superpixel Network:* Unlike traditional superpixel segmentation networks constructed from a single set of superpixels, the ASSN is designed to integrate high-dimensional and complex spectral information together with spatial edge information into the segmentation process. It further applies the differentiable simple linear iterative clustering (SLIC) algorithm, enabling iterative optimization via a loss function and thereby overcoming the limitation of traditional methods that cannot improve segmentation quality through feedback. The generation of HSI-compliant superpixels provides a solid foundation for the GCN to efficiently capture comprehensive global spatial information, ensuring that the segmentation is directly guided by classification tasks.

The ASSN is designed with two dedicated feature extraction channels: the spatial edge channel and the spectral channel. The spatial edge channel first applies PCA for dimensionality reduction, followed by the Sobel operator to detect spatial edges. The spectral channel is implemented as a multilayer CNN that captures multiscale spectral patterns through upsampling and downsampling operations. Then, the spectral

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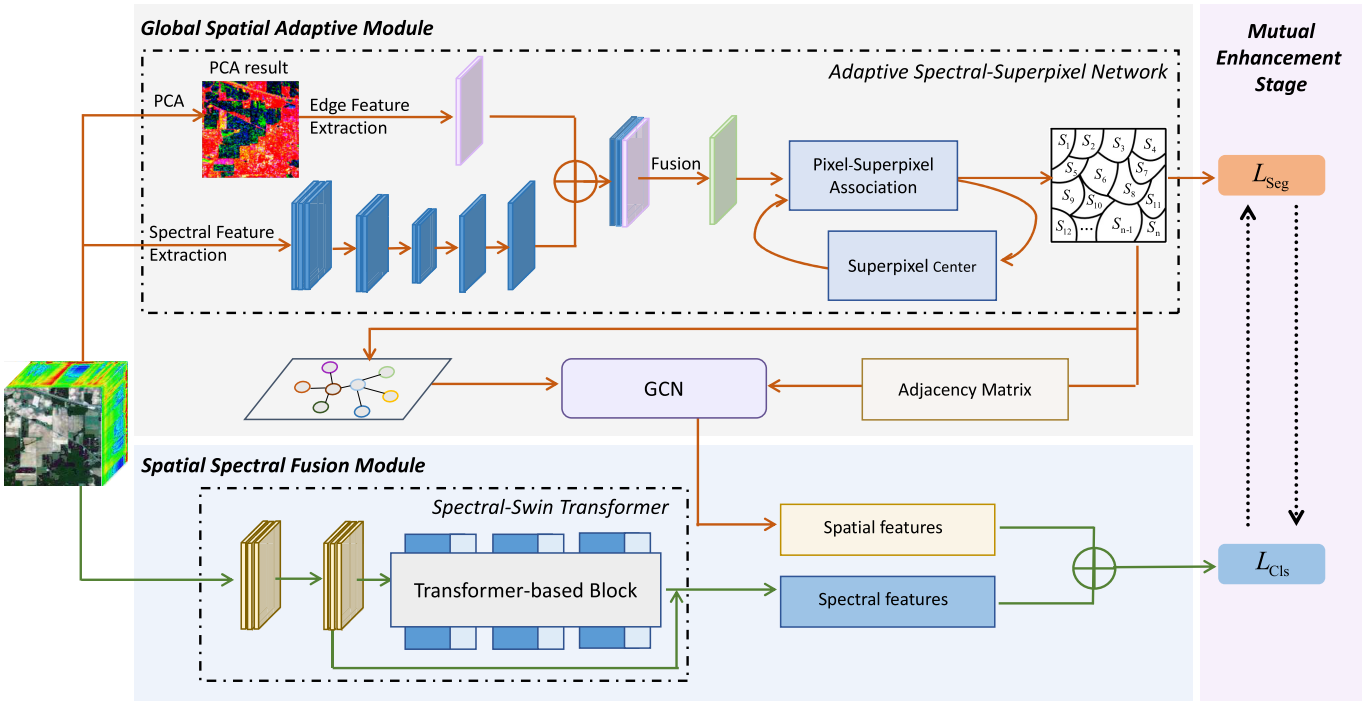



Fig. 2. Overall architecture of the proposed $S^2$CMEN. The framework consists of three main components: 1) the GSAM, which captures global spatial dependencies using an ASSN and a GCN; 2) the SSFM, which incorporates a Spectral-Swin Transformer to extract long-range detial spectral features; and 3) a mutual enhancement mechanism, where segmentation and classification branches are jointly optimized via a unified loss. Arrows indicate the direction of feature flow between modules.

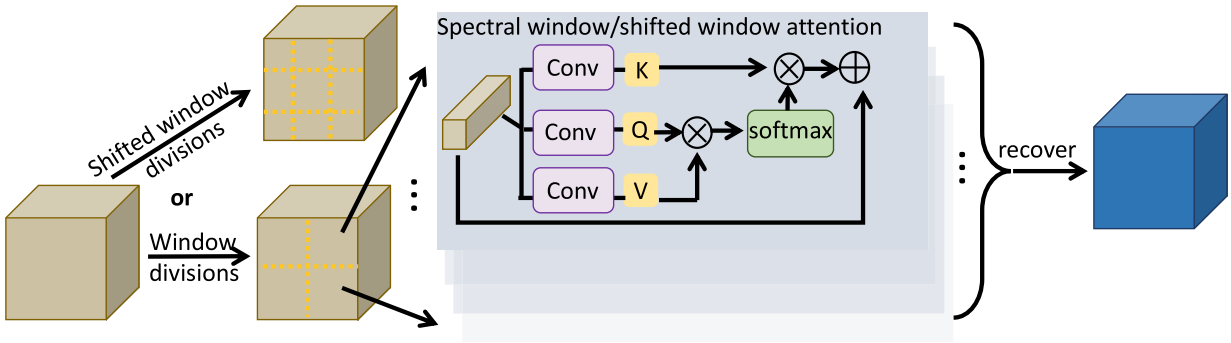



Fig. 3. Illustration of the spectral window attention (SW-Attention) and spectral window shift attention (SWS-Attention) mechanisms in the Spectral-Swin Transformer. To better process high-dimensional spectral features, we introduce convolutional operations during the computation of the query (Q), key (K), and value (V) matrices.

information and spatial information are fused by combining each pixel's spectral vector with its spatial coordinates. The resulting spectral–spatial representation is then processed by a shallow convolutional block, which enhances feature coherence and suppresses noise in both the spectral and spatial domains.

Finally, the refined features are fed into a differentiable SLIC module. Guided by a learnable similarity metric, each pixel is softly assigned to multiple superpixel centers, enabling the segmentation to adapt to the intrinsic spectral and spatial structure of the image. Consequently, the ASSN produces superpixels that preserve both spectral similarity and spatial continuity, accurately delineating object boundaries in HSIs and providing a robust foundation for subsequent global spatial modeling in the GSAM.

The main design of the ASSN is inspired by the superpixel sampling network [39]. We adopt soft associations $U \in \mathbb{R}^{hw\times 9}$, restricted to the nine surrounding superpixels, to replace hard nearest neighbor assignments with differentiable local operations. This allows pixel–superpixel relationships to be learned and updated during training, thereby enhancing the adaptability of the segmentation. Formally, given an HSI $I \in \mathbb{R}^{hw\times b}$, for a pixel $p$ and a superpixel $i$ at iteration $t$, the pixel–superpixel association in differentiable SLIC can be expressed as

$$U_{pi}^{t} = e^{-D(I_p, S_i^{t-1})} = e^{-\|I_p - S_i^{t-1}\|^2} \tag{1}$$

where $D$ denotes the distance computation $D(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|^2$.

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Accordingly, the computation of new superpixel cluster centers is adjusted to be the weighted sum of pixel features

$$S_i^t = \frac{1}{Z_i^t} \sum_{p=1}^{n} U_{pi}^t I_p \tag{2}$$

where $Z_i^t = \sum_p U_{pi}^t$ is the normalization constant.

Given the microscopic nature of this superpixel algorithm, we can leverage deep feature extractors and train the entire network, eliminating the need for manually designed pixel features.

In addition, to ensure the stability of the association relation, we construct an unweighted adjacency mask $\mathbf{M}$ as

$$\mathbf{M}_{i,j}^t = \begin{cases} 1, & \text{if } \mathcal{S}_i^t \text{ and } \mathcal{S}_j^t \text{ are adjancent} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\mathbf{M}_{i,j}^t$ is the value of $\mathbf{M}$ at position $(i, j)$ during iteration $t$ and $\mathcal{S}_i^t$ and $\mathcal{S}_i^t$ are adjacent when they share a boundary.

*2) GCN:* We use graphs to describe one-to-many relationships in non-Euclidean space in HSI. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph, where $\mathcal{V}$ and $\mathcal{E}$ denote the vertex set and the edge set, respectively. In practice, $\mathcal{V}$ and $\mathcal{E}$ are generally encoded as a node matrix $\mathbf{H}$ and an adjacency matrix $\mathbf{A}$.

In order to work with an adaptable ASSN, we learn the edge weights through the GCN itself. To save computation and get an improved feature representation, Kipf and Welling [41] add the unit matrix $\mathbf{F}$ to the adjacency matrix $\mathbf{A}$. Liu et al. [27] propose to compute the edge weights from the input features of each layer, and letting $\mathbf{H}^l$ be the output of the $l$th graph convolutional layer, the adjacency matrix $\mathbf{A}^l$ can be expressed as

$$\mathbf{A}^l = \text{Sigmoid}\left(\left(\tilde{\mathbf{H}}^{l-1}\mathbf{W}_\phi^l\right)\left(\tilde{\mathbf{H}}^{l-1}\mathbf{W}_\phi^l\right)^{\mathrm{T}}\right) \odot \mathbf{M} + \mathbf{F} \tag{4}$$

where $\tilde{\mathbf{H}}^{l-1}$ denotes the normalized input feature map of the $l$th graph convolutional layer, $\mathbf{W}_\phi$ is the parameter to be learned, and the operator $\odot$ denotes the Hadamard product. The GCN operation formula is as follows:

$$\mathbf{H}^l = f\left(\left(\mathbf{D}^l\right)^{-\frac{1}{2}} \mathbf{A}^l \left(\mathbf{D}^l\right)^{-\frac{1}{2}} \tilde{\mathbf{H}}^{l-1}\mathbf{W}^l\right) \tag{5}$$

where $\mathbf{D}^l$ is the degree matrix of $\mathbf{A}^l$ and $\mathbf{W}^l$ is the parameter to be learned.

In order to fully extract the global spatial features processed through superpixel segmentation, we set the number of GCN layers to 2.

Notably, GSAM can function as a plug-and-play spatial enhancement module, in which the captured global spatial information from HSI is used to strengthen the spatial representation capabilities of various spectral feature extractors, such as 2D-CNNs, Swin Transformers, and other commonly used networks in HSI tasks. By injecting global spatial context into these models, GSAM is expected to alleviate the locality limitations that are often inherent in spectral-dominant methods.

To further demonstrate the versatility and effectiveness of GSAM as a generic plug-in, we quantitatively evaluate its impact on spatial–spectral representation learning and classification performance by integrating it into several mainstream HSI classification backbones, as presented in Section IV.

### C. Spatial–Spectral Fusion Module

In this module, the obtained global spatial and long-range detail spectral features are fused for classification. Global spatial features are obtained from GSAM, whereas the extraction of spectral features mainly depends on the designed Spectral-Swin Transformer.

*1) Spectral-Swin Transformer:* The Spectral-Swin Transformer first uses two convolutional layers for mapping features and inputs them to the Transformer-based block. Based on the idea of Swin Transformer, these blocks split the original large-size features into different windows through the window-shifting mechanism and calculate the attention of each window separately. This operation cannot only reduce the amount of computation but also can effectively extract the long-range detail spectral features. The partitioned windows are not the same as patches, which are independent inputs, but the window-shifting mechanism can be partitioned by different windows. This makes the attention not isolated but can be combined through different window segmentation to achieve the global interaction of the spectral information.

These blocks are composed of spectral window attention (SW-Attention) and spectral window-shifting attention (SWS-Attention). These efforts reduce computational overhead and ensure effective information interaction between two different windows. A LayerNorm (LN) layer is added between SW-Attention and SWS-Attention to make the training more stable, and a remaining connection is used after each module. This part can be expressed as follows:

$$\hat{z}^l = SW - \text{Attention}\left(LN\left(z^{l-1}\right)\right) + z^{l-1} \tag{6}$$

$$\hat{z}^{l+1} = SWS - \text{Attention}\left(LN\left(z^l\right)\right) + z^l \tag{7}$$

where $\hat{z}^l$ and $z^l$ denote the output features of the SW-Attention, SWS-Attention, respectively.

*SW-Attention and SWS-Attention:* As illustrated in Fig. 3, the SW-Attention module constrains the self-attention computation to nonoverlapping local windows, significantly improving computational efficiency while maintaining strong performance, even with large-scale HSI inputs. To better handle the high-dimensional spectral characteristics of HSI, we incorporate convolutional operations in the extraction of the Q, K, and V matrices, thereby enhancing the spectral self-attention capability.

Following SW-Attention, the SWS-Attention module extends the standard window-based self-attention by applying a cyclic window shift prior to the attention computation. Specifically, after SW-Attention partitions the feature map into regular nonoverlapping windows, SWS-Attention shifts these windows by half the window size along both spatial dimensions (height and width). This shifted partitioning allows each pixel to interact with its counterparts in neighboring windows, thus establishing cross-window dependencies that are not captured in standard window-based self-attention.

While performing global self-attention across the entire image is computationally prohibitive for HSI data, this offset window mechanism effectively balances the tradeoff between efficiency and contextual modeling. Compared to traditional attention mechanisms, the combination of SW-Attention and

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SWS-Attention not only captures fine-grained local spectral details but also models broader spatial–spectral correlations. By alternating between the two modules, the SSFM block ensures dense and diverse spectral feature coverage across multiple layers of shifted windows, thereby significantly enhancing the model's global representational capacity for HSI.

As SW-Attention and SWS-Attention need to cooperate to ensure effective feature extraction, we set the number of Transformer-based blocks in the Spectral-Swin Transformer to 2. This configuration ensures efficient feature extraction while avoiding the overfitting phenomenon caused by excessive training of Transformer-based blocks.

The Spectral-Swin Transformer is meticulously designed to adeptly manage the complexities posed by large HSI inputs. Simultaneously, it excels in capturing local spectral information. Through the local–global joint extraction of spectral information, the Spectral-Swin Transformer proficiently acquires long-range detail spectral features.

### D. Mutual Enhancement Stage

In the MES, the network jointly optimizes the superpixel segmentation branch and the classification branch in a unified framework. The segmentation branch focuses on learning discriminative superpixel representations, while the classification branch aims to predict class probabilities for each pixel. Both branches provide feedback to each other during training, enabling mutual improvement.

For the superpixel segmentation branch, in order to better learn a specific superpixel representation, we use the semantic label $T$ in HSI as the pixel attribute representation. The representation resulting $\bar{T}$ using row-normalized association matrix $\hat{U}$, $\bar{T} = \hat{U}S$. Then, the superpixel segmentation loss is given as

$$L_{\text{Seg}} = L\left(T, \bar{T}\right) = L\left(T, \hat{U}S\right). \tag{8}$$

For the classification branch, in order to generate the probability of each class on each pixel, we classify the feature mapping $F$ using a softmax classifier to get the predicted value $\hat{T}$. The network is finally trained using the cross-entropy loss

$$L_{\text{Cls}} = L\left(T, \hat{T}\right). \tag{9}$$

In order to integrate superpixel segmentation and classification into a unified network, we design a joint task loss, in which two tasks interact and update the network simultaneously. This interaction aims to utilize the strengths of each task to improve the overall performance. The weighted computation of the segmentation loss $L_{\text{Cls}}$ and classification loss $L_{\text{Seg}}$ constitutes the mutual loss $L_{\text{mutual}}$. The process is as follows:

$$L_{\text{mutual}} = \alpha L_{\text{Seg}} + (1-\alpha) L_{\text{Cls}} \tag{10}$$

where $\alpha$ is a learnable parameter used to balance the contributions of the segmentation task and the classification task. By allowing the learnable parameter to adaptively adjust, the model is able to better find the optimal loss combination in complex multitask learning scenarios, thus improving the performance of the two tasks and promoting synergy.

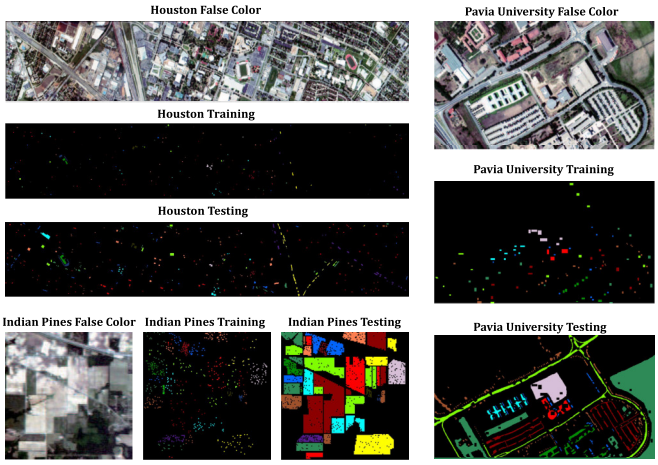



Fig. 4. False-color images and the distribution of training and test sets on the three well-known datasets, i.e., IP, PU, and Houston.

With the interaction of the two loss functions, the classification and the superpixel segmentation are mutually reinforcing. More accurate superpixel results provide richer global spatial information for HSI classification. The enhancement of classification in turn facilitates the updating of the superpixel centers, making the results of superpixel segmentation closer to the true distribution of the classes.

## IV. Experiments

We conduct experiments on three well-known HSI datasets. First, we introduce the datasets and experimental setup, including implementation details and the state-of-the-art methods used for comparison. Then, we carry out comprehensive comparative tests to demonstrate the superiority of our proposed method over other backbones. Finally, we perform extensive experiments with ablation analysis to evaluate the critical design elements of $\mathbf{S}^2$CMEN.

### A. Datasets Description

We selected three datasets for the experiments, namely, Indian Pines (IPs) [42], Pavia University (PU),[1] and Houston 2013 (Houston).[2] Tables I–III provide the detailed sample information for each dataset. Fig. 4 displays the three datasets' false-color HSI and the spatial distribution of the training and testing samples.

*1) IP Dataset:* The initial HSI dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). It covers northwestern Indiana, USA, and comprises $145 \times 145$ pixels with a ground sampling distance (GSD) of 20 m. The dataset includes 200 spectral bands (excluding 20 noisy and water absorption bands, i.e., 104–108, 150–163, and 220) ranging from 400 to 2500 nm, with a spectral resolution of 10 nm.

*2) PU Dataset:* The second HSI dataset was captured over PU and its surroundings in Pavia, Italy, by the reflective optics system imaging spectrometer (ROSIS). The image consists of

[1]Pavia scenes were provided by Prof. Paolo Gamba from the Telecommunications and Remote Sensing Laboratory, PU (Italy).

[2]The data were provided by Prof. N. Y Okoya from the University of Tokyo and RIKEN AIP.

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TABLE I

LAND-COVER CLASSES OF THE IP DATASET, WITH THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

| Class NO. | Class Name | Training | Testing |
|---|---|---|---|
| 1 | Corn Notill | 50 | 1384 |
| 2 | Corn Mintill | 50 | 784 |
| 3 | Corn | 50 | 184 |
| 4 | Grass Pasture | 50 | 447 |
| 5 | Grass Trees | 50 | 694 |
| 6 | Hay Windrowed | 50 | 439 |
| 7 | Soybean Notill | 50 | 918 |
| 8 | Soybean Mintill | 50 | 2418 |
| 9 | Soybean Clean | 50 | 564 |
| 10 | Wheat | 50 | 162 |
| 11 | Woods | 50 | 1244 |
| 12 | Buildings Grass Trees Drives | 50 | 330 |
| 13 | Stone Steel Towers | 50 | 45 |
| 14 | Alfalfa | 15 | 39 |
| 15 | Grass Pasture Mowed | 15 | 11 |
| 16 | Oats | 15 | 5 |
| | total | 695 | 9671 |

TABLE II

LAND-COVER CLASSES OF THE PU DATASET, WITH THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

| Class NO. | Class Name | Training | Testing |
|---|---|---|---|
| 1 | Asphalt | 548 | 6304 |
| 2 | Meadows | 540 | 18146 |
| 3 | Gravel | 392 | 1815 |
| 4 | Trees | 524 | 2912 |
| 5 | Metal Sheets | 265 | 1113 |
| 6 | Bare Soil | 532 | 4572 |
| 7 | Bitumen | 375 | 981 |
| 8 | Bricks | 514 | 3364 |
| 9 | Shadows | 231 | 795 |
| | total | 3921 | 40002 |

TABLE III

LAND-COVER CLASSES OF THE HOUSTON DATASET, WITH THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

| Class NO. | Class Name | Training | Testing |
|---|---|---|---|
| 1 | Healthy Grass | 198 | 1384 |
| 2 | Stress Grass | 190 | 784 |
| 3 | Synthetic Grass | 192 | 184 |
| 4 | Tree | 188 | 447 |
| 5 | Soil | 186 | 694 |
| 6 | Water | 182 | 439 |
| 7 | Residential | 196 | 918 |
| 8 | Commercial | 191 | 2418 |
| 9 | Road | 193 | 564 |
| 10 | Highway | 191 | 162 |
| 11 | Railway | 181 | 1244 |
| 12 | Parking Lot1 | 192 | 330 |
| 13 | Parking Lot2 | 184 | 45 |
| 14 | Tennis Court | 181 | 39 |
| 15 | Running Track | 187 | 11 |
| | total | 2832 | 12197 |

$610 \times 340$ pixels with a GSD of 1.3 m and records 103 spectral bands ranging from 430 to 860 nm.

TABLE IV

PARAMETRIC EXPERIMENTS FOR THE NUMBER OF SUPERPIXELS PRESET FOR SUPERPIXEL SEGMENTATION ON THREE DATASETS IN OA

| Number | Matrix | | |
|---|---|---|---|
| | IP | PU | Houston |
| 80 | 97.01 | 91.94 | 90.71 |
| 100 | 97.15 | **92.33** | **91.38** |
| 120 | 97.32 | 91.73 | 90.88 |
| 140 | **97.38** | 91.75 | 90.86 |

*3) Houston Dataset:* The third dataset was captured by the ITRES CASI-1500 sensor over the University of Houston campus and its surrounding rural areas in Texas, USA. As a widely used benchmark for land-cover classification, the hyperspectral cube consists of $349 \times 1905$ pixels with 144 wavelength bands ranging from 364 to 1046 nm at 10-nm intervals. Notably, the Houston 2013 dataset used is a cloud-free version, processed to recover missing data or remove occlusions using illumination-related threshold maps.

### B. Experimental Setup

The networks discussed in this article are implemented using the PyTorch platform. We employ an "exponential" learning rate policy, dynamically updating the current learning rate by multiplying a base learning rate by $5e^{-5}$ every 50 epochs, and the Adam optimizer is used for network optimization. During training, the maximum number of epochs is set to 200. Batch normalization (BN) with a momentum of 0.9 is applied, and the batch size is set to 64. For models requiring patch input, a patch size of $5 \times 5$ is specified.

We quantitatively evaluate each model's classification performance using three common indices: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. Additionally, we visualize the classification maps from different models for qualitative comparison.

### C. Parametric Analysis

*1) Number of Superpixels Preset:* We analyze the number of superpixels preset for superpixel segmentation, which is used to validate the effect of superpixel results on our method. As shown in Table IV, the optimal preset values for superpixel segmentation vary across different datasets. For the IP dataset, due to the large number of classes and their relatively dense distribution, the best results are achieved with a preset value of 140. More comprehensive superpixel segmentation enhances the capture of global spatial information for this dataset. In contrast, the PU and Houston datasets exhibit different characteristics: the PU dataset has relatively fewer classes, and the Houston dataset has a sparser distribution of samples. Consequently, an excessive number of superpixels can lead to an overemphasis on mixed spatial information. Thus, for both the PU and Houston datasets, a preset value of 100 yields more prominent results.

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TABLE V
QUANTITATIVE PERFORMANCE OF VARIOUS CLASSIFICATION METHODS ON THE IP DATASET, INCLUDING OA, AA, KAPPA, AND CLASS-SPECIFIC ACCURACIES

| CLASS NO. | 2-D CNN | miniGCN | Swin Transformer | SpectralFormer | SSFTT | morphFormer | GraphGST | 3DSS-Mamba | Our |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 69.58 | 72.54 | 66.76 | 67.19 | 92.10 | 81.06 | 91.18 | 86.21 | **92.80** |
| 2 | 65.05 | 55.99 | 67.47 | 68.49 | 92.70 | 85.71 | **98.72** | 90.66 | 96.81 |
| 3 | 84.23 | 92.93 | 87.50 | 93.47 | **100.00** | **100.00** | 99.46 | 93.91 | **100.00** |
| 4 | 93.28 | 92.62 | 91.27 | 90.38 | 87.02 | 96.86 | 96.20 | 94.78 | **97.09** |
| 5 | 88.66 | 94.98 | 91.67 | 90.24 | **100.00** | 99.85 | 95.70 | 95.13 | **100.00** |
| 6 | 97.03 | 98.63 | 97.03 | 96.35 | 98.63 | 99.31 | 98.63 | 90.26 | 98.41 |
| 7 | 75.59 | 64.71 | 70.69 | 69.06 | 75.38 | 93.24 | **95.10** | 48.00 | 92.96 |
| 8 | 73.69 | 68.78 | 75.80 | 75.68 | 84.55 | 90.77 | 98.35 | **98.40** | 94.85 |
| 9 | 84.75 | 69.33 | 80.31 | 74.82 | 80.14 | 84.39 | 94.28 | 93.21 | **95.66** |
| 10 | 99.38 | 98.77 | 98.76 | 99.38 | **100.00** | 99.38 | **100.00** | 96.77 | 98.14 |
| 11 | 81.67 | 87.78 | 95.73 | 94.29 | 87.86 | 97.82 | 91.12 | **98.16** | 97.33 |
| 12 | 68.18 | 50.00 | 53.03 | 76.06 | 90.30 | 98.78 | 98.48 | 89.54 | **100.00** |
| 13 | 85.15 | **100.00** | 95.56 | **100.00** | **100.00** | 97.77 | **100.00** | 93.98 | **100.00** |
| 14 | 89.74 | 78.72 | 84.62 | 85.36 | **100.00** | 97.43 | **100.00** | 99.55 | **100.00** |
| 15 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | 96.22 | **100.00** |
| 16 | **100.00** | 80.00 | 60.00 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** |
| OA | 77.97 | 75.48 | 79.02 | 81.42 | 95.66 | 91.74 | 96.85 | 95.67 | **97.38** |
| AA | 85.52 | 86.36 | 82.86 | 85.13 | 93.04 | 95.15 | 97.34 | 92.87 | **97.74** |
| Kappa | 0.7486 | 0.7486 | 0.7594 | 0.8086 | 0.9612 | 0.9052 | 0.9612 | 95.07 | **0.9672** |

TABLE VI
QUANTITATIVE PERFORMANCE OF VARIOUS CLASSIFICATION METHODS ON THE PU DATASET, INCLUDING OA, AA, KAPPA, AND CLASS-SPECIFIC ACCURACIES

| CLASS NO. | 2-D CNN | miniGCN | Swin Transformer | SpectralFormer | SSFTT | morphFormer | GraphGST | 3DSS-Mamba | Our |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 84.72 | **96.35** | 71.51 | 85.48 | 80.18 | 90.13 | 81.88 | 88.21 | 93.76 |
| 2 | 88.32 | 89.42 | 76.82 | 63.18 | **96.66** | 95.67 | 93.75 | 94.47 | 92.87 |
| 3 | 76.08 | 87.01 | 46.39 | 75.86 | 80.99 | 74.26 | 78.78 | 73.16 | **88.04** |
| 4 | 95.94 | 94.26 | 96.39 | 77.35 | 92.99 | 96.53 | **97.87** | 95.98 | 96.35 |
| 5 | 99.37 | 99.82 | 99.19 | 99.82 | 99.82 | 99.82 | **100.00** | 99.64 | 99.82 |
| 6 | 66.25 | 43.12 | 83.18 | **98.31** | 73.81 | 80.35 | 93.96 | 80.35 | 78.93 |
| 7 | 85.32 | 90.88 | 83.08 | 94.03 | 86.03 | 91.13 | 89.50 | 91.84 | **95.31** |
| 8 | 94.23 | 77.42 | 89.03 | 93.57 | 96.04 | **97.50** | 95.65 | 97.77 | 92.47 |
| 9 | 94.96 | 87.27 | 96.48 | **97.35** | 95.84 | 97.10 | 94.96 | 97.35 | 96.60 |
| OA | 86.10 | 80.79 | 82.59 | 91.08 | 90.24 | 91.75 | 91.78 | 89.20 | **92.33** |
| AA | 87.25 | 85.07 | 80.22 | 87.05 | 89.16 | 91.82 | 91.82 | 88.81 | **92.69** |
| Kappa | 0.8138 | 0.7877 | 0.7923 | 0.8433 | 0.8676 | 0.8892 | 0.8904 | 0.8834 | **0.8963** |

### *D. Quantitative Results and Analysis*

For comparison, several representative baseline methods have been selected, including 2-D CNN [43], miniGCN [22], Swin Transformer [13], SpectralFormer [28], SSFTT [44], morphFormer [29], GraphGST [45], and 3DSS-Mamba [46]. Tables IV–VI present the classification results for three HSI datasets in OA, AA, and Kappa, respectively. The accuracies for each class are also shown in these three tables.

The 2-D CNN and miniGCN exhibit strong performance because of their powerful learning capabilities. However, CNNs focus excessively on labels of local spatial information and require more labeled training datasets for accurate classification, which is contrary to the HSI classification task. miniGCN, although it can obtain spatial information, is not capable of handling spectral sequence information. Moreover, neither CNN nor GCN-based methods are good at handling spectral details, which makes their performance poor in overall classification.

Transformer-based methods such as Swin Transformer, SpectralFormer, SSFTT, morphFormer, GraphGST, and 3DSS-Mamba extract finer global spectral information, achieving better performance than CNN or GCN-based methods. However, SpectralFormer has a barrier in extracting local information, which is not conducive to fully extracting local and global fusion information. Others, except GraphGST, can excel in obtaining local information, but they lack the ability to extract global spatial features. For GraphGST, the use of absolute position can enhance the representation learning capability of the Transformer. However, without the addition of global spatial features, the performance is still slightly inferior. 3DSS-Mamba, as an efficient feature extraction method, combines the advantages of long-range dependency modeling and linear computational complexity. However, it still lacks the ability to effectively capture global contextual information and fine-grained edge details. In summary, the Transformer-based backbone gets good results to different degrees but

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TABLE VII
QUANTITATIVE PERFORMANCE OF VARIOUS CLASSIFICATION METHODS ON THE HOUSTON DATASET, INCLUDING OA, AA, KAPPA, AND CLASS-SPECIFIC ACCURACIES

| CLASS NO. | 2-D CNN | miniGCN | Swin Transformer | SpectralFormer | SSFTT | morphFormer | GraphGST | 3DSS-Mamba | Our |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 82.71 | **98.39** | 82.80 | 87.08 | 84.42 | 82.81 | 83.01 | 92.26 | 84.80 |
| 2 | 98.40 | 92.11 | 96.62 | 95.30 | 97.27 | 97.27 | 99.34 | 98.05 | **99.81** |
| 3 | 93.66 | 99.60 | 99.80 | 98.41 | 95.84 | 98.41 | 95.84 | 91.27 | **100.00** |
| 4 | 96.49 | 96.78 | 99.24 | 92.32 | 97.53 | 98.86 | 98.67 | 93.11 | **99.71** |
| 5 | **100.00** | 97.73 | 97.73 | **100.00** | **100.00** | 99.24 | **100.00** | 99.24 | **100.00** |
| 6 | 92.30 | 95.10 | 95.10 | 95.10 | 95.10 | 96.50 | 97.90 | 92.30 | **100.00** |
| 7 | 86.47 | 57.28 | 76.99 | 80.87 | 89.73 | **90.85** | 89.17 | 88.21 | 88.43 |
| 8 | 77.87 | 68.09 | 55.55 | 77.68 | 77.01 | 86.51 | **86.70** | 85.49 | 84.42 |
| 9 | 72.52 | 53.92 | 67.42 | **89.70** | 78.78 | 82.53 | 82.81 | 88.42 | 82.90 |
| 10 | 50.96 | 77.47 | 68.05 | 57.33 | 64.28 | 58.39 | 70.94 | 78.31 | **89.57** |
| 11 | 72.86 | 84.91 | 82.35 | 85.76 | 82.92 | 82.63 | **92.88** | 89.54 | 86.71 |
| 12 | 66.18 | 77.23 | 58.50 | 89.91 | 83.38 | 88.76 | 78.10 | 89.22 | **90.01** |
| 13 | 75.78 | 50.88 | 60.00 | 88.42 | **91.92** | 91.22 | 90.52 | 87.53 | 87.36 |
| 14 | 92.71 | 98.38 | 98.79 | **100.00** | 97.97 | 98.78 | 98.78 | **100.00** | **100.00** |
| 15 | 96.61 | 98.52 | 98.73 | 97.04 | 94.08 | **99.36** | 97.46 | 98.67 | 93.86 |
| OA | 81.99 | 81.71 | 80.41 | 88.14 | 86.86 | 88.28 | 89.28 | 88.94 | **91.38** |
| AA | 83.71 | 83.09 | 82.50 | 89.32 | 88.68 | 90.51 | 90.81 | 91.44 | **92.21** |
| Kappa | 0.805 | 0.8018 | 0.8497 | 0.8626 | 0.8573 | 0.8728 | 0.8836 | 0.8945 | **0.9065** |

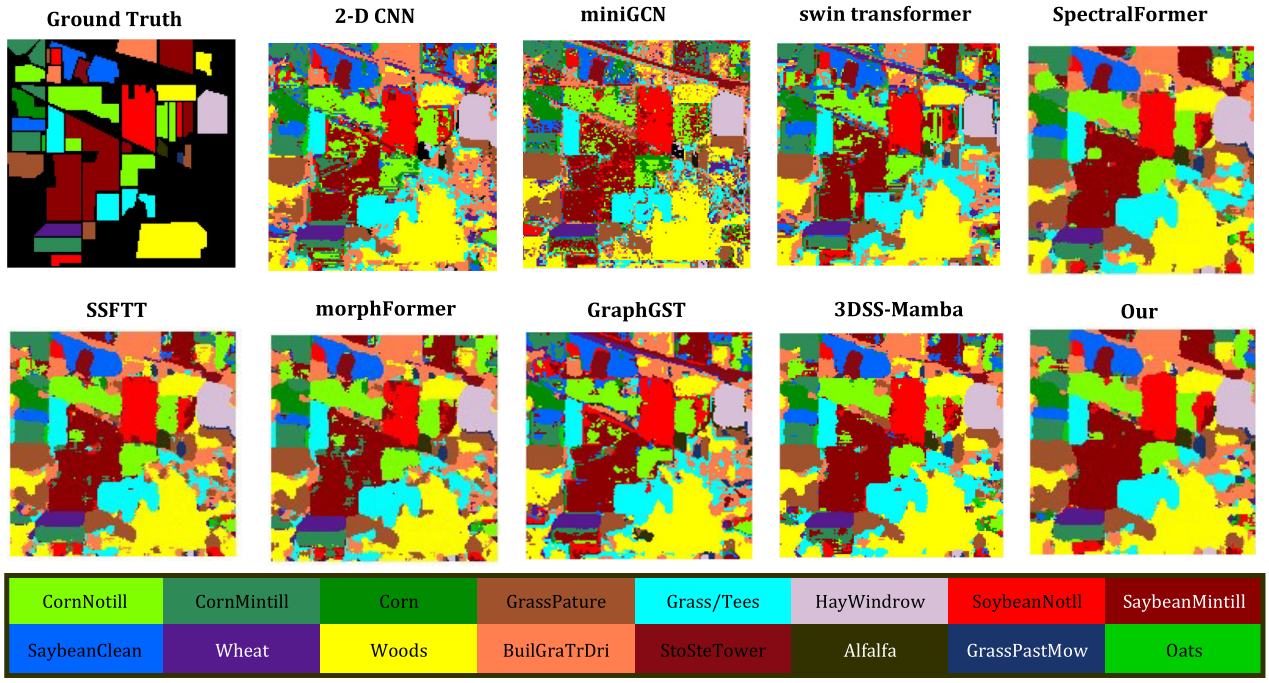



Fig. 5. Classification maps obtained by different models on the IP dataset.

has their own limitations, which are eventually shown in the classification results.

The proposed $S^2$CMEN consistently outperforms existing classification methods, achieving superior accuracy across all evaluated datasets. This is due to its ability to acquire global spatial and long-range detail spectral features, resulting in rich visual representations. In addition, on some unbalanced and limited training samples in the IP dataset, $S^2$CMEN can utilize the window-shifting mechanism to perform sufficient window interactions to distinguish the detailed features of similar analogs and obtain good classification results.

We qualitatively evaluate the classification methods by visualizing the classification maps. Figs. 5–7 show the results for the IP, PU, and Houston datasets, respectively. DL-based methods, including CNNs and GCNs, generate relatively smooth classification maps thanks to their robust ability to model nonlinear datasets. Transformer-based methods achieve relatively ideal classification maps with well-preserved texture and edge details by aggregating spectral information. Notably, our proposed $S^2$CMEN reveals more realistic and finer details. The advantage of incorporating global spatial information is its ability to more accurately discriminate among limited samples

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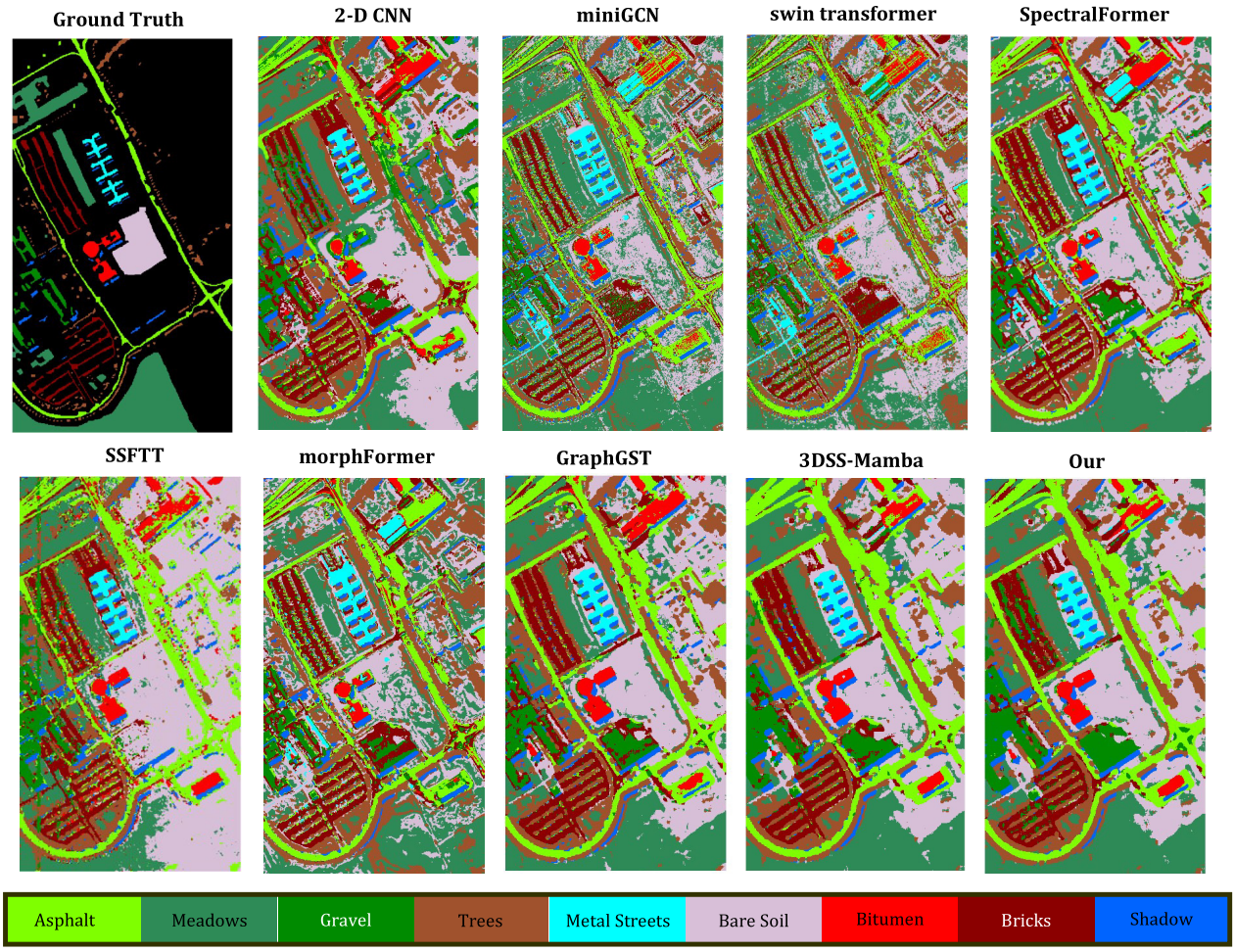



Fig. 6. Classification maps obtained by different models on the PU dataset.

or imbalanced classes (such as in the IP datasets), resulting in a clearer classification map. This observation intuitively demonstrates the effectiveness and superiority of the proposed method.

### *E. Ablation Experiments*

*1) GSAM:* In order to demonstrate the effectiveness of GSAM, we use three datasets for ablation experiments to investigate the performance of the module in HSI classification by combining it with different backbones.

We selected two representative baselines (2-D CNN and Swin Transformer), which are capable of processing large-size images, to evaluate the performance of the GSAM module across three datasets. In these experiments, we employed a feature-level concatenation strategy to fuse the global spatial features extracted by GSAM with the spectral features extracted by each baseline. Specifically, the spatial and spectral features are concatenated along the channel dimension to form a joint representation. This concatenated feature vector is then passed through a lightweight fusion module comprising two fully connected layers with a ReLU activation in between. The module compresses the joint representation back to the original embedding dimension, thereby achieving spatial–spectral feature fusion. It can be seen in Fig. 8 that the lowest classification accuracy is obtained by directly applying the Swin Transformer because the Swin Transformer has an advantage on local–global spectral features but ignores the extraction of global spatial features. This may reduce the classification accuracy. Swin Transformer adds GSAM to improve the accuracy, but the effect is limited. The 2-D CNN has a different degree of improvement after adding GSAM. This suggests that the inclusion of global spatial information can be of great help in HSI classification. However, they still neglect the extraction of long-range detail spectral features. Our $\mathbf{S}^2$CMEN achieves the best results with GSAM. This is because the designed SSFM can extract sufficiently long-range detail spectral features. Also, it is matched with the global spatial information from GSAM to obtain the desired results.

Taken together, the effectiveness of GSAM is most evident on the IP dataset due to its wide and dense classes. The challenge of accurately classifying HSI without focusing on global spatial information is mitigated by the addition of GSAM, which significantly improves the classification accuracy. In addition, although the PU and Houston datasets have fewer classes or more scattered class distributions, the

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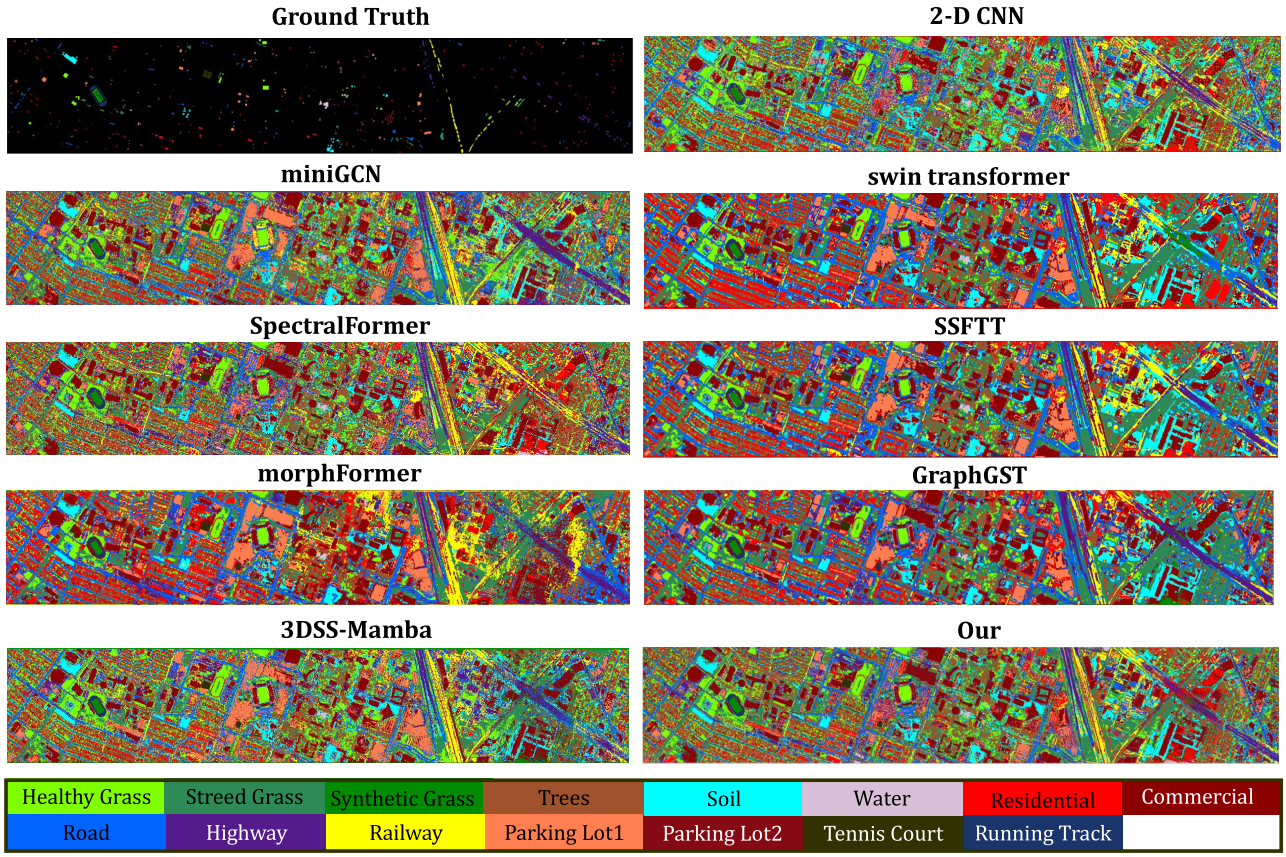



Fig. 7. Classification maps obtained by different models on the Houston dataset.

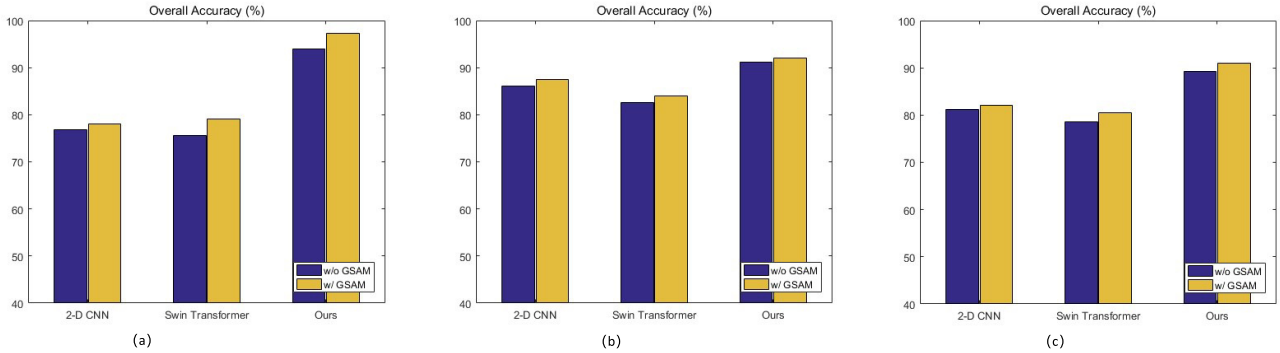



Fig. 8. Ablation experiments obtained by GSAM on three datasets. (a) IP dataset. (b) PU dataset. (c) Houston dataset.

integration of GSAM still yields improvements to varying degrees.

*2) Spectral-Swin Transformer:* We also validate the effectiveness of the designed Spectral-Swin Transformer. In order to achieve the demand of matching large-size HSI spectral information extraction, we developed Spectral-Swin Transformer, which is capable of extracting long-range detail spectral features. To indicate the Spectral-Swin Transformer's validity, we compare it with the Swin Transformer and experiment with the Spectral-Swin Transformer by adding and subtracting convolution mapping layers in three datasets. As shown in Table VIII, it can be seen that Swin Transformer is difficult to complete the requirement of spectral detail extraction in HSI, and its performance is mediocre. The performance of Spectral-Swin Transformer without convolution mapping layers is improved, which shows that our improvement of SW/SWS-Attention mechanism is effective. However, the lack of convolution mapping layers is still a bottleneck in processing large-size HSI. The full Spectral-Swin Transformer performs best. This shows that it is able to fully extract the multiscale information. With such a convolution mapping layer's effort, the Spectral-Swin Transformer finally acquires long-range detail spectral features for matching global spatial information.

*3) Mutual Enhancement Stage:* To verify the effectiveness of the mutual promotion between superpixel segmentation and classification, we conducted experiments on three datasets, comparing the performance of using only the

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TABLE VIII
ABLATION EXPERIMENTS OBTAINED BY SPECTRAL-SWIN TRANSFORMER ON THREE DATASETS IN OA

| Moudle | Matrix | | |
|---|---|---|---|
| | IP | PU | Houston |
| Swin Transformer | 79.02 | 82.59 | 80.41 |
| Spectral-Swin Transformer w/o mapping layers | 93.25 | 88.86 | 86.42 |
| Spectral-Swin Transformer | **96.32** | **91.23** | **88.95** |

TABLE IX
ABLATION EXPERIMENTS ON THREE DATASETS USING DIFFERENT COMBINATIONS OF LOSS FUNCTIONS IN OA

| loss function | Matrix | | |
|---|---|---|---|
| | IP | PU | Houston |
| Classification Loss Only | 96.19 | 91.95 | 89.74 |
| Joint Classification + Superpixel Segmentation loss | 97.38 | 92.33 | 91.38 |

classification loss versus the proposed joint loss that integrates both classification and segmentation tasks, as shown in Table IX.

It can be observed that the classification accuracy drops when using only the classification loss. This is because, in the absence of segmentation guidance, the superpixel segmentation process struggles to adaptively refine the clustering centers, leading to unspecific and less informative segmentation results. As a consequence, the global features extracted by GSAM are followed by inaccurate clustering, which negatively affects the classification performance.

In contrast, the proposed joint loss facilitates mutual guidance between classification and segmentation. The segmentation process, being task-adaptive, generates more meaningful and structure-aware superpixels, which in turn enhance classification accuracy, particularly at class boundaries, demonstrating the effectiveness of the MES.

## V. CONCLUSION

HSI classification plays a vital role in remote sensing applications. While patch-based models achieve competitive results, they often neglect the broader correlations embedded in global spatial information. To address this limitation, we propose $\mathbf{S}^2$CMEN, a novel framework that mutually enhances superpixel segmentation and classification to improve the extraction of global spatial dependencies and long-range spectral details. Specifically, $\mathbf{S}^2$CMEN first captures global spatial features through the GSAM, which is designed as a plug-and-play module to enhance other spectral extractors as well. SSFM then employs an improved Spectral-Swin Transformer to model long-range spectral dependencies, which are subsequently fused with the global spatial information. Finally, a novel MES enables segmentation and classification to mutually guide each other, optimizing overall network performance.

Despite its effectiveness, $\mathbf{S}^2$CMEN has certain limitations. First, the dual-branch architecture may limit its applicability in real-time or resource-constrained scenarios. Second, the performance of the GSAM module is partially dependent on the quality of the superpixel segmentation, which can be affected by complex textures or spectral noise in real-world HSI data. Third, although the model demonstrates good generalization across three datasets, its robustness to variations in sensor types and environmental conditions still requires further investigation. In future work, we plan to address these limitations by exploring advanced strategies, such as improved superpixel generation methods, enhanced GCN architectures, and the development of lightweight Transformer-based modules. These improvements aim to further optimize the model's efficiency, robustness, and applicability for real-world HSI classification tasks.

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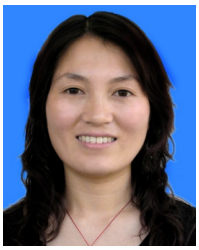
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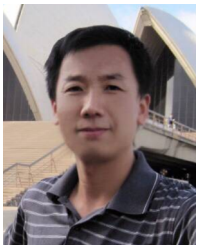
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